A Technical Report for Polyglot-Ko: Open-Source Large-Scale Korean Language Models

Hyunwoong Ko*, Kichang Yang*, Minho Ryu*, Taekyoon Choi*
Seungmu Yang*, Jiwung Hyun*, Sungho Park*, Kyubyong Park†

EleutherAI

arXiv:2306.02254v2 [cs.CL] 6 Jun 2023

Abstract

Polyglot is a pioneering project aimed at enhancing the non-English language performance of multilingual language models. Despite the availability of various multilingual models such as mBERT (Devlin et al., 2019), XGLM (Lin et al., 2022), and BLOOM (Scao et al., 2022), researchers and developers often resort to building monolingual models in their respective languages due to the dissatisfaction with the current multilingual models' non-English language capabilities. Addressing this gap, we seek to develop advanced multilingual language models that offer improved performance in non-English languages. In this paper, we introduce the Polyglot Korean models, which represent a specific focus rather than being multilingual in nature. In collaboration with TUNiB[1], our team collected 1.2TB of Korean data meticulously curated for our research journey. We made a deliberate decision to prioritize the development of Korean models before venturing into multilingual models. This choice was motivated by multiple factors: firstly, the Korean models facilitated performance comparisons with existing multilingual models; and finally, they catered to the specific needs of Korean companies and researchers. This paper presents our work in developing the Polyglot Korean models, which propose some steps towards addressing the non-English language performance gap in multilingual language models.

1 Introduction

The advent of large-scale language models has revolutionized the field of natural language processing, leading to significant advancements in various applications, such as language translation and text classification (Devlin et al., 2019; Radford et al., 2019; Liu et al., 2019; Clark et al., 2020; Chowdhery et al., 2022; Anil et al., 2023). While numerous large language models for English have been publicly released (Zhang et al., 2022; Black et al., 2022; Biderman et al., 2023; Touvron et al., 2023; Computer, 2023; Team, 2023), the availability of such models for non-English languages remains limited. Although several multilingual large language models have also been released (Lin et al., 2022; Scao et al., 2022), they are typically trained on English-centric corpora, resulting in lower performance on other languages.

However, with the increasing interest in non-English languages, there is a growing need for high-performance language models that are specifically tailored to these languages. To tackle this challenge, we initiated the Polyglot project, which focuses on the development of large language models customized for non-English languages. As part of the project, our first model in this endeavor is Polyglot-Ko, an exceptional language model specifically designed for the Korean language. We chose to prioritize the Korean language as the initial model because our founding members are primarily Korean and we had a readily available dataset for training purposes. Our goal is to make these language models accessible to researchers and practitioners, empowering them to explore and advance natural language processing tasks in their respective languages. Polyglot-Ko leverages the transformer architecture, known for its effectiveness in capturing long-range dependencies in natural language text. Our model has been trained on an extensive corpus of text data, incorporating diverse sources such as web pages, news articles, and social media posts to ensure its versatility across different domains and styles.

In this technical report, we provide a comprehensive description of the architecture and training process for four distinct Polyglot-Ko models, differing in parameter sizes: 1.3B (billion), 3.8B,

* Equal contribution
† Affiliation: TUNiB
[1] https://tunib.ai/

5.8B, and 12.8B. Notably, our 12.8 billion parameter model represents the largest publicly available Korean language model suitable for commercial applications, making it an invaluable resource for researchers and practitioners engaged in Korean natural language processing tasks.

We assess the zero-shot and few-shot performance of our Polyglot-Ko models using the KOBEST benchmark (Kim et al., 2022). Through our experiments, we have successfully demonstrated that Polyglot-Ko attains competitive results across various benchmark datasets.

In addition to presenting our achievements, we also acknowledge potential limitations and identify areas that warrant future improvement. By offering recommendations for further research, we aspire to foster advancements in the field. We firmly believe that Polyglot-Ko will serve as a valuable resource for the Korean natural language processing community, enabling the development of innovative applications and contributing to a deeper understanding of the intricacies and dynamics of the Korean language.

2 Datasets

We collaborated with TUNiB to collect a large-scale Korean language dataset for our research. The dataset, totaling 1.2TB, was meticulously gathered through our collaborative efforts. Subsequently, we performed preprocessing on this dataset, resulting in 863GB of text data that served as the foundation for our analysis and model training.

Source	Size (GB)
Korean blog posts	682.3
Korean news dataset	87.0
Modu corpus	26.4
Korean patent dataset	19.0
Korean Q&A dataset	18.1
KcBert dataset	12.7
Korean fiction dataset	6.1
Korean online comments	4.2
Korean wikipedia	1.4
Clova call	< 1.0
Naver Sentiment Movie Corpus	< 1.0
Korean hate speech dataset	< 1.0
Open subtitles	< 1.0
AIHub various tasks datasets	< 1.0
Standard Korean dictionary	< 1.0

Table 1: Datasets for the Korean language.

2.1 Data Analysis

The primary motivation behind our data analysis is to mitigate potential risks that may arise during both training and inference stages. We strive to overcome various issues that can adversely impact model performance and reliability. For instance, we identify and handle challenges such as empty or excessively short text data, repeated words and characters, and instances of duplicated data, which can pose problems during model training. Additionally, we pay particular attention to the inference stage, where the presence of personally identifiable information (PII) can create complications. By thoroughly analyzing the data, we strive to minimize these risks and ensure the robustness and privacy compliance of our models. Through a meticulous examination of the data with the aim of mitigating these risks, we successfully categorized them into four distinct types:

- **Data available for training**: This category predominantly comprises news and Wikipedia data, which provide substantial information with sufficiently long text sequences.
- **Data requiring contextual information for training**: In this category, we primarily encountered blog data and news data. These datasets contained numerous short texts that were incorrectly scraped, necessitating the inclusion of contextual information during the training process.
- **Data containing hate speech**: We observed a significant presence of hate speech in datasets sourced from certain community websites, highlighting the importance of addressing this issue during model training.
- **NLP task-specific data**: This category encompasses data specifically designed for NLP tasks such as text classification or entity recognition. While this data can be utilized for model training, it necessitates separate handling during model evaluation.

During the examination of data types, we encountered various quality issues that required preprocessing to ensure optimal learning. Recognizing the potential detrimental impact of these issues on model training, we organized them and incorporated them into our text preprocessing pipeline. The specific quality issues we addressed include:

- **Empty text**: Instances with no text content.
- **Unnecessary spaces**: Instances containing multiple unnecessary spaces.
- **De-identification**: Identification and removal of personally identifiable information within the data instances.
- **Uncleaned HTML tags**: Removal of HTML tags that had not been properly cleaned.
- **Deduplication**: Identification and removal of duplicated data instances based on exact matches.
- **Broken code**: Handling instances where only fragments of HTML or Markdown were present.
- **Short text**: Detection and handling of excessively short data instances.
- **Repeated characters**: Identification and addressing of instances with repeated characters.

By addressing these quality issues as part of our text preprocessing workflow, we aimed to enhance the quality and reliability of the data used for model training. One crucial aspect of our data preprocessing was the removal of HTML-like code to a large extent, as the focus of the polyglot model is on generating Korean text. Additionally, the length of the data played a significant role in our preprocessing efforts. Longer text lengths provide more contextual information for model training, while shorter text lengths limit the available context for text generation.

3 Models

To train the polyglot models, we used EleutherAI's GPT-NeoX codebase (Andonian et al., 2021). This codebase provided a solid foundation for our training process. Additionally, we received invaluable assistance from Stability AI[2], who provided access to 256 A100s (8 * 32 nodes) on HPC clusters[3]. This computational infrastructure played a crucial role in efficiently training our models. The training tokens information for each model is as follows:

- The 1.3B model was trained on 213B tokens.
- The 3.8B model was trained on 219B tokens.
- The 5.8B model was trained on 172B tokens.
- The 12.8B model was trained on 167B tokens.

Due to the limitations of available computational resources, the models were trained with varying numbers of tokens. Despite our efforts [4], moreover, broken generation (e.g generate same tokens repeatedly) occurred as the loss sharply dropped near the 1 epoch boundary for 1.3B and 3.8B models. Consequently, we made the decision to select and early stop the model checkpoints prior to the epoch boundary, as they exhibited better generation performance.

A consistent tokenizer was utilized across all models, featuring a vocabulary size of 30003. The tokenizer was trained using morpheme-aware Byte-Level BPE (Byte-Pair Encoding). We performed morpheme analysis using MeCab[5], a widely-used morphological analysis tool for Korean text. This ensured that the models were equipped with an effective and consistent tokenization scheme tailored to the Korean language.

Now, let's delve into the details of each model's training process:

1.3B Model The 1.3B model was trained without model parallelism, utilizing a total batch size of 1024. However, broken generation was observed as the loss sharply dropped around the 100,000 steps. To handle this, model checkpoints were evaluated and verified prior to the onset of it to ensure optimal model selection.

3.8B Model Similar to the 1.3B model, the 3.8B model also experienced same symptoms at around 100,000 steps. Model parallelism was applied during training, and the overall batch size remained the same as that of the 1.3B model. As a result, the decision was made to halt the model training process.

5.8B Model The 5.8B model utilized model parallelism, and the overall batch size was reduced by 1/4 compared to the 1.3B and 3.8B models. We trained with 172B tokens, which consisted of a total of 320,000 steps. As it was trained with lower tokens compared with 1.3B and 3.8B, The model's performance consistently improved as the number of training steps increased.

[2] https://stability.ai/
[3] https://hpc.stability.ai/
[4] We suspect it was overtrained or overfitting, but there are not any sciencific evidence.
[5] https://bitbucket.org/eunjeon/mecab-ko-dic/src/master/

Hyperparameter	**1.3B**	**3.8B**	**5.8B**	**12.8B**
$n_{parameters}$	1,331,810,304	3,809,974,272	5,885,059,072	12,898,631,680
n_{layers}	24	32	28	40
d_{model}	2,048	3,072	4,096	5,120
d_{ff}	8,192	12,288	16,384	20,480
n_{heads}	16	24	16	40
d_{head}	128	128	256	128
n_{vocab}	30,003 / 30,080	30,003 / 30,080	30,003 / 30,080	30,003 / 30,080
Positional Encoding	Rotary (RoPE)	Rotary (RoPE)	Rotary (RoPE)	Rotary (RoPE)
RoPE Dimensions	64	64	64	64

Table 2: The configuration settings of the Polyglot-Ko model.

12.8B Model For the 12.8B model, model parallelism was employed, with a scale of 2 times larger than the 5.8B model. The overall batch size was maintained through the use of gradient accumulation steps (GAS). The model was trained for a total of 301,000 steps.

4 Experiments

We conducted an evaluation of Polyglot-Ko on the KOBEST dataset (Kim et al., 2022), which encompasses five downstream tasks: COPA, HellaSwag, SentiNeg, BoolQ, and WiC. Each task focuses on different aspects of language understanding and reasoning.

- **COPA** requires the selection of an alternative that is a cause/effect of a given premise.
- **HellaSwag** evaluates commonsense reasoning and natural language inference.
- **BoolQ** is designed to test the models' ability to answer questions that require reasoning over multiple sentences.
- **SentiNeg** focuses on sentiment analysis in the Korean language.
- **WiC** requires identifying whether the meaning of a target word is the same or different in two given contexts.

During the evaluation, we compared the performance of Polyglot-Ko with other similar models, including ko-gpt-trinity-1.2B[6], KoGPT (Kim et al., 2021), and XGLM-7.5B (Lin et al., 2022). These models were chosen as they are currently the only publicly available billion-scale Korean language models, excluding other multilingual models. The evaluation process involved using the provided prompts, and F1-scores were used as the evaluation metric for all tasks. We utilized the polylgot[7] branch of EleutherAI's lm-evaluation-harness (Gao et al., 2021) repository, which provided the evaluation codebase for our experiments. This allowed for consistent and standardized evaluation procedures across different models, ensuring fair comparisons.

Note that we have chosen not to report the performance of Polyglot-Ko on the WiC task in this section due to nearly random performance observed across all models. However, detailed results for the WiC task can be found in the Appendix B. Figure 1 illustrates the results obtained when varying the number of few-shot examples.

4.1 COPA

The experiment was conducted on the COPA task, and the results are presented in the left portion of Table 3. The table provides a comprehensive comparison of model performance based on the number of few-shot examples. To ensure fairness, all models were evaluated under the same conditions and using identical prompts. The results clearly demonstrate that our 12.8B model outperforms the other models across all scenarios. Specifically, our 12.8B model achieves the highest F1 score in both 0-shot and 50-shot settings. For 0-shot, our model achieves an impressive F1 score of 0.7937, surpassing all other models. Similarly, in the 50-shot scenario, our 12.8B model achieves the highest F1 score of 0.8368. These results highlight the superiority of our 12.8B model over comparable models

[6] https://huggingface.co/skt/ko-gpt-trinity-1.2B-v0.5

[7] https://github.com/EleutherAI/lm-evaluation-harness/tree/polyglot

Model	Params	COPA (n=shot)				HellaSwag			
		n=0	n=5	n=10	n=50	n=0	n=5	n=10	n=50
Ko-GPT-Trinity	1.2B	0.670	0.648	0.642	0.651	0.524	0.527	0.517	0.535
Polyglot-Ko (ours)	1.3B	0.720	0.719	0.720	0.721	0.525	0.526	0.528	0.543
Polyglot-Ko (ours)	3.8B	0.760	0.761	0.764	0.779	0.571	0.583	0.567	0.579
Polyglot-Ko (ours)	5.8B	0.775	0.768	0.778	0.789	0.598	0.600	0.598	**0.621**
KoGPT	6.0B	0.735	0.729	0.728	0.748	0.559	0.583	0.583	0.591
xGLM	7.5B	0.672	0.673	0.677	0.712	0.566	0.569	0.556	0.562
Polyglot-Ko (ours)	12.8B	**0.794**	**0.811**	**0.804**	**0.837**	**0.595**	**0.631**	**0.610**	0.612

Table 3: Comparative results of the performance (F1 score) on the COPA and HellaSwag tasks.

Model	Params	BoolQ (n=shot)				SentiNeg			
		n=0	n=5	n=10	n=50	n=0	n=5	n=10	n=50
Ko-GPT-Trinity	1.2B	0.336	0.401	0.364	0.356	0.606	0.688	0.728	0.841
Polyglot-Ko (ours)	1.3B	0.355	0.475	0.411	0.404	0.679	0.626	0.551	0.785
Polyglot-Ko (ours)	3.8B	0.432	0.526	0.493	0.404	0.486	0.795	0.732	0.785
Polyglot-Ko (ours)	5.8B	0.436	0.570	0.519	0.524	0.339	0.884	0.881	0.952
KoGPT	6.0B	0.451	0.598	0.550	0.520	0.375	0.894	0.929	0.970
xGLM	7.5B	0.446	0.332	0.332	0.332	0.358	0.447	0.396	0.527
Polyglot-Ko (ours)	12.8B	**0.482**	**0.604**	**0.629**	**0.645**	**0.912**	**0.902**	**0.934**	**0.972**

Table 4: Comparative results of the performance (F1 score) on the BoolQ and SentiNeg tasks.

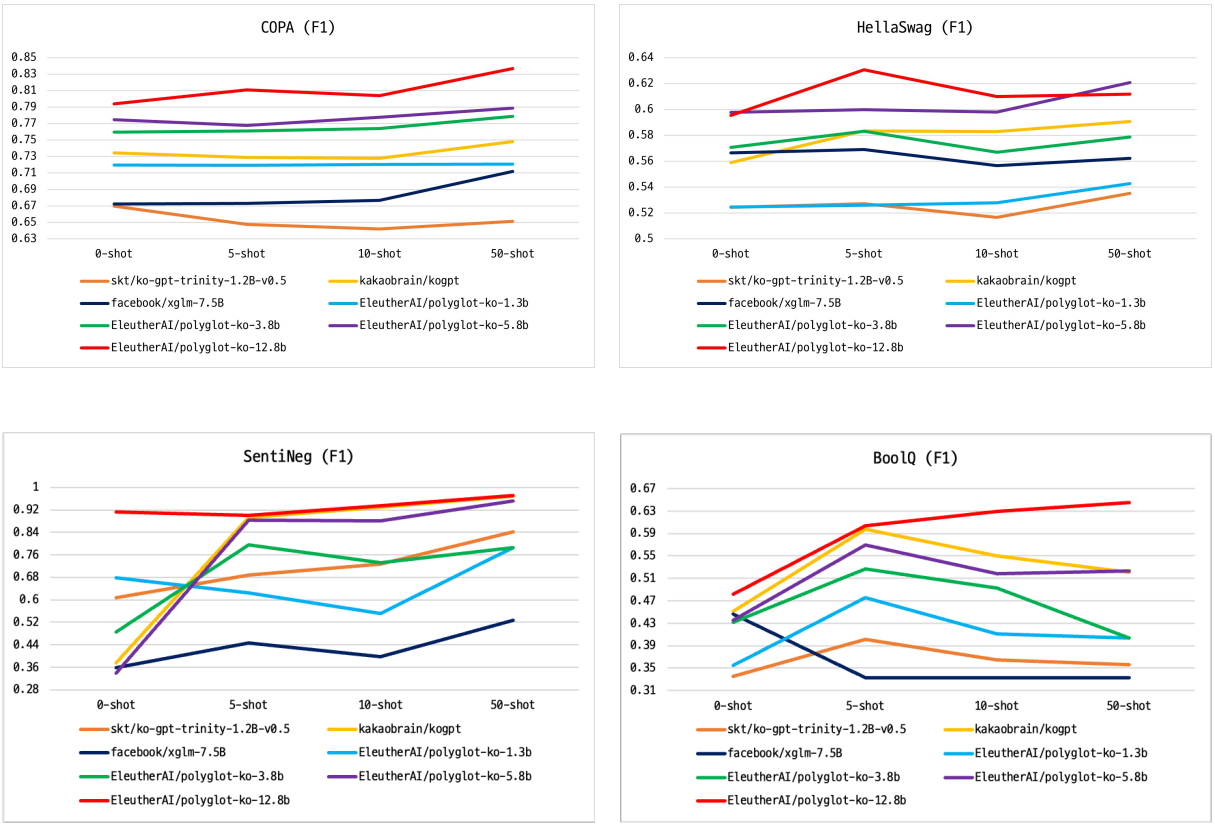

Figure 1: The performance metrics of COPA (top left), HellaSwag (top right), SentiNeg (bottom left), and BoolQ (bottom right) tasks using the KOBEST dataset, with all metrics measured using the F1 score.

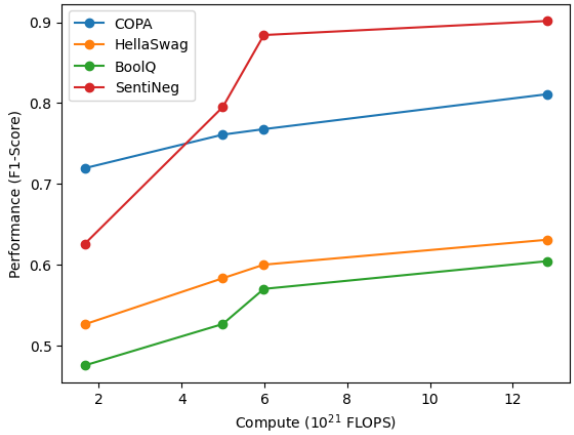

Figure 2: The 5-shot performance of Polyglot-Ko models on each task demonstrates a clear trend that as the compute increases, the performance improves.

in the COPA task. It showcases the model's exceptional performance and reinforces its effectiveness in understanding and reasoning tasks.

4.2 HellaSwag

The right portion of Table 3 presents the performance of different language models on the HellaSwag task for the Korean language. The table includes the model name, the number of parameters in each model, and their respective performance at different number of shots on the HellaSwag task using the KOBEST dataset. The evaluation metric used is the F1 score. According to the table, our 12.8B model achieves the highest scores for the 5-shot, 10-shot, and 50-shot scenarios, with F1 scores of 0.6306, 0.6098, and 0.6118, respectively. However, in the 0-shot scenario, kakaobrain's kogpt model achieves the highest score of 0.5590. Overall, our 12.8B model demonstrates the best performance among the listed models on the HellaSwag task for the Korean language. These results highlight the effectiveness of our 12.8B model in understanding and generating coherent responses within the context of the HellaSwag task. Its superior performance in the majority of the shot scenarios showcases its ability to handle complex language understanding tasks and generate contextually appropriate responses.

4.3 BoolQ

In the BoolQ task, which focuses on answering boolean questions, a thorough analysis of the results demonstrates that our models outperformed the others. Specifically, our largest model, Polyglot-ko-12.8B, achieved the highest F1 scores. This indicates that the model possesses exceptional accuracy in predicting answers to boolean questions. Conversely, SKT's ko-gpt-trinity model exhibited relatively lower F1 scores across all prompt numbers, while Facebook's XGLM model consistently underperformed. As a result, our models demonstrate strong performance in the BoolQ task.

4.4 SentiNeg

The SentiNeg task results are also presented in the left portion of Table 4. This task focuses on sentiment analysis for negation detection. A comprehensive analysis of the results reveals that our models exhibited superior performance. In particular, our 12.8B model achieved the highest F1 scores, indicating its excellent ability to accurately detect negated sentiment. SKT's ko-gpt-trinity model also showed consistent improvement, while kakaobrain's KoGPT model demonstrated varied performance with slight increases or decreases in F1 scores depending on the prompt number. However, Facebook's XGLM model consistently displayed lower performance in the SentiNeg task.

Furthermore, during our investigation, we discovered that the default prompt used for this task introduced significant instability, particularly in zero-shot performance. To address this issue, we devised a modified prompt, which led to substantial improvements in the model's performance. For detailed results, please refer to the Appendix. Overall, our models demonstrate strong performance in the SentiNeg task, showcasing their effectiveness in sentiment analysis and negation detection.

When analyzing the performance of Polyglot-Ko models only in the 5-shot evaluation across all four tasks, it becomes clear that the performance improves with increasing compute(Kaplan et al., 2020) as depicted in Figure 2.

5 Limitations and Disclaimers

Polyglot-Ko has been primarily trained to optimize next token prediction, which makes it suitable for a wide range of tasks. However, it is crucial to acknowledge the potential for unexpected outcomes. While Polyglot-Ko strives to generate the most statistically likely response, it may not always provide the most accurate or factual answer. It is important to exercise caution when relying on the model's outputs.

Additionally, it is worth noting that Polyglot-Ko

may generate content that is socially unacceptable or offensive. To mitigate this risk, we strongly recommend implementing a human curator or employing other filtering mechanisms to censor sensitive or inappropriate content. Regarding the hardware used for training, it is important to mention that the models were trained on a hardware setup with relatively low TFLOPS compared to the upcoming versions of Polyglot that we are currently preparing. This resulted in longer training time and resources to complete the training process successfully.

Furthermore, we discovered mistakes in the data preprocessing phase during our experiments. Specifically, the data was incorrectly stripped of newlines, leading to a loss of document structure. This likely resulted in some information loss during the model training process. It is important to address this issue in future iterations to ensure the preservation of document structure and minimize information loss. These considerations highlight the importance of continuously improving the training process and addressing any limitations or errors that arise. By doing so, we can enhance the performance and reliability of Polyglot-Ko for a wide range of tasks and applications.

6 Conclusion

Currently, we are actively working on training the new version of the Polyglot Korean language model. Our aim is to expand its capacity to eventually reach 40B parameters. This process has involved significant trial and error as we strive to enhance the performance and capabilities of the model.

Based on our experience and expertise in developing Korean language models, we have also embarked on the creation of two types of multilingual models. The first type is an East-Asian model, which includes Korean, Chinese, Japanese, Indonesian, Malay, Vietnamese, Thai, and English. This model aims to cater to the linguistic needs of countries in the East-Asian region.

The second type is a Romance model, which incorporates Spanish, Portuguese, French, Romanian, and Italian. This model is designed to support the linguistic requirements of Romance language-speaking countries.

By developing these multilingual models, we aim to democratize and promote access to language model technology across the globe. We believe that this will contribute to the advancement of research and academics in various countries, allowing users to leverage the power of language models for diverse applications and linguistic contexts. We are excited about the potential impact and benefits these models can bring to researchers, practitioners, and language enthusiasts worldwide.

Acknowledgements

We are grateful to Stability AI for generously providing the essential computing resources that were instrumental in the successful execution of this project. Their support and infrastructure were crucial for training and evaluating our models.

Additionally, we would like to extend our appreciation to TUNiB for their invaluable contribution in providing a large-scale Korean dataset. This dataset played a pivotal role in the development and training of our language models, and we are grateful for their collaboration and partnership.

Finally, we would like to thank Stella Biderman for her valuable feedback on the paper, which greatly enhanced the quality and clarity of our work.

References

Alex Andonian, Quentin Anthony, Stella Biderman, Sid Black, Preetham Gali, Leo Gao, Eric Hallahan, Josh Levy-Kramer, Connor Leahy, Lucas Nestler, Kip Parker, Michael Pieler, Shivanshu Purohit, Tri Songz, Wang Phil, and Samuel Weinbach. 2021. GPT-NeoX: Large Scale Autoregressive Language Modeling in PyTorch.

Rohan Anil, Andrew M. Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, Eric Chu, Jonathan H. Clark, Laurent El Shafey, Yanping Huang, Kathy Meier-Hellstern, Gaurav Mishra, Erica Moreira, Mark Omernick, Kevin Robinson, Sebastian Ruder, Yi Tay, Kefan Xiao, Yuanzhong Xu, Yujing Zhang, Gustavo Hernandez Abrego, Junwhan Ahn, Jacob Austin, Paul Barham, Jan Botha, James Bradbury, Siddhartha Brahma, Kevin Brooks, Michele Catasta, Yong Cheng, Colin Cherry, Christopher A. Choquette-Choo, Aakanksha Chowdhery, Clément Crepy, Shachi Dave, Mostafa Dehghani, Sunipa Dev, Jacob Devlin, Mark Díaz, Nan Du, Ethan Dyer, Vlad Feinberg, Fangxiaoyu Feng, Vlad Fienber, Markus Freitag, Xavier Garcia, Sebastian Gehrmann, Lucas Gonzalez, Guy Gur-Ari, Steven Hand, Hadi Hashemi, Le Hou, Joshua Howland, Andrea Hu, Jeffrey Hui, Jeremy Hurwitz, Michael Isard, Abe Ittycheriah, Matthew Jagielski, Wenhao Jia, Kathleen Kenealy, Maxim Krikun, Sneha Kudugunta, Chang Lan, Katherine Lee, Benjamin Lee, Eric Li, Music Li, Wei Li, YaGuang Li,

Jian Li, Hyeontaek Lim, Hanzhao Lin, Zhongtao Liu, Frederick Liu, Marcello Maggioni, Aroma Mahendru, Joshua Maynez, Vedant Misra, Maysam Moussalem, Zachary Nado, John Nham, Eric Ni, Andrew Nystrom, Alicia Parrish, Marie Pellat, Martin Polacek, Alex Polozov, Reiner Pope, Siyuan Qiao, Emily Reif, Bryan Richter, Parker Riley, Alex Castro Ros, Aurko Roy, Brennan Saeta, Rajkumar Samuel, Renee Shelby, Ambrose Slone, Daniel Smilkov, David R. So, Daniel Sohn, Simon Tokumine, Dasha Valter, Vijay Vasudevan, Kiran Vodrahalli, Xuezhi Wang, Pidong Wang, Zirui Wang, Tao Wang, John Wieting, Yuhuai Wu, Kelvin Xu, Yunhan Xu, Linting Xue, Pengcheng Yin, Jiahui Yu, Qiao Zhang, Steven Zheng, Ce Zheng, Weikang Zhou, Denny Zhou, Slav Petrov, and Yonghui Wu. 2023. Palm 2 technical report.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling.

Sid Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, USVSN Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. 2022. Gpt-neox-20b: An open-source autoregressive language model.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, Kensen Shi, Sasha Tsvyashchenko, Joshua Maynez, Abhishek Rao, Parker Barnes, Yi Tay, Noam Shazeer, Vinodkumar Prabhakaran, Emily Reif, Nan Du, Ben Hutchinson, Reiner Pope, James Bradbury, Jacob Austin, Michael Isard, Guy Gur-Ari, Pengcheng Yin, Toju Duke, Anselm Levskaya, Sanjay Ghemawat, Sunipa Dev, Henryk Michalewski, Xavier Garcia, Vedant Misra, Kevin Robinson, Liam Fedus, Denny Zhou, Daphne Ippolito, David Luan, Hyeontaek Lim, Barret Zoph, Alexander Spiridonov, Ryan Sepassi, David Dohan, Shivani Agrawal, Mark Omernick, Andrew M. Dai, Thanumalayan Sankaranarayana Pillai, Marie Pellat, Aitor Lewkowycz, Erica Moreira, Rewon Child, Oleksandr Polozov, Katherine Lee, Zongwei Zhou, Xuezhi Wang, Brennan Saeta, Mark Diaz, Orhan Firat, Michele Catasta, Jason Wei, Kathy Meier-Hellstern, Douglas Eck, Jeff Dean, Slav Petrov, and Noah Fiedel. 2022. Palm: Scaling language modeling with pathways.

Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020. Electra: Pre-training text encoders as discriminators rather than generators.

Together Computer. 2023. Redpajama: An open source recipe to reproduce llama training dataset.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding.

Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, Jason Phang, Laria Reynolds, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. 2021. A framework for few-shot language model evaluation.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models.

Dohyeong Kim, Myeongjun Jang, Deuk Sin Kwon, and Eric Davis. 2022. Kobest: Korean balanced evaluation of significant tasks.

Ildoo Kim, Gunsoo Han, Jiyeon Ham, and Woonhyuk Baek. 2021. Kogpt: Kakaobrain korean(hangul) generative pre-trained transformer. https://github.com/kakaobrain/kogpt.

Xi Victoria Lin, Todor Mihaylov, Mikel Artetxe, Tianlu Wang, Shuohui Chen, Daniel Simig, Myle Ott, Naman Goyal, Shruti Bhosale, Jingfei Du, Ramakanth Pasunuru, Sam Shleifer, Punit Singh Koura, Vishrav Chaudhary, Brian O'Horo, Jeff Wang, Luke Zettlemoyer, Zornitsa Kozareva, Mona Diab, Veselin Stoyanov, and Xian Li. 2022. Few-shot learning with multilingual language models.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. OpenAI blog, 1(8):9.

Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, Matthias Gallé, et al. 2022. Bloom: A 176b-parameter open-access multilingual language model. arXiv preprint arXiv:2211.05100.

MosaicML NLP Team. 2023. Introducing mpt-7b: A new standard for open-source, ly usable llms. Accessed: 2023-03-28.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. Llama: Open and efficient foundation language models.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. 2022. Opt: Open pre-trained transformer language models.

A Prompt Modification for SentiNeg Task

In the case of SentiNeg, we observed that the prompt used in the KoBEST paper employed a simple classification task. However, the range of questions encompassed a wide spectrum and often exhibited ambiguity. Consequently, we made arbitrary modifications to the prompt in order to obtain results. The results of these prompt modifications are presented in the left portion of Table 5 and Figure 3a. It is evident from the results that the scores achieved with the modified prompt are substantially higher on average than those achieved with the original prompt. This demonstrates the importance of prompt design and customization in achieving better performance in the SentiNeg task.

B Results for the WiC task

The results of the WiC task are presented in the right portion of Table 5 and visualized in Figure 3b. Notably, all models demonstrated random performance in this task. We utilized the accuracy metric as it provided a more straightforward evaluation of performance, particularly for random performance. The consistent random performance across all models indicates that they encountered challenges in making accurate predictions for the WiC task. Therefore, further investigation and improvements are necessary to enhance the models' capability to effectively tackle this task.

Model	Params	**Modified SentiNeg** (n=shot)				**WiC**			
		n=0	n=5	n=10	n=50	n=0	n=5	n=10	n=50
Ko-GPT-Trinity	1.2B	0.767	0.699	0.713	0.836	0.487	0.492	0.479	0.475
koGPT	6B	0.927	0.955	0.955	0.952	0.484	0.495	0.479	0.475
xGLM	7.5B	0.835	0.745	0.797	0.727	0.488	0.490	0.498	**0.514**
Polyglot-Ko (ours)	1.3B	0.889	0.850	0.887	0.907	0.489	0.486	**0.506**	0.487
Polyglot-Ko (ours)	3.8B	**0.942**	0.894	0.906	0.952	0.489	**0.499**	0.491	0.488
Polyglot-Ko (ours)	5.8B	0.878	0.927	0.906	0.960	0.484	0.494	0.480	0.479
Polyglot-Ko (ours)	12.8B	0.893	**0.985**	**0.982**	**0.982**	**0.493**	0.494	0.488	0.487

Table 5: Comparative results of the performance on the SentiNeg task with a modified prompt and the WiC task, with the F1 scores for SentiNeg while the accuracy for WiC to show random performance.

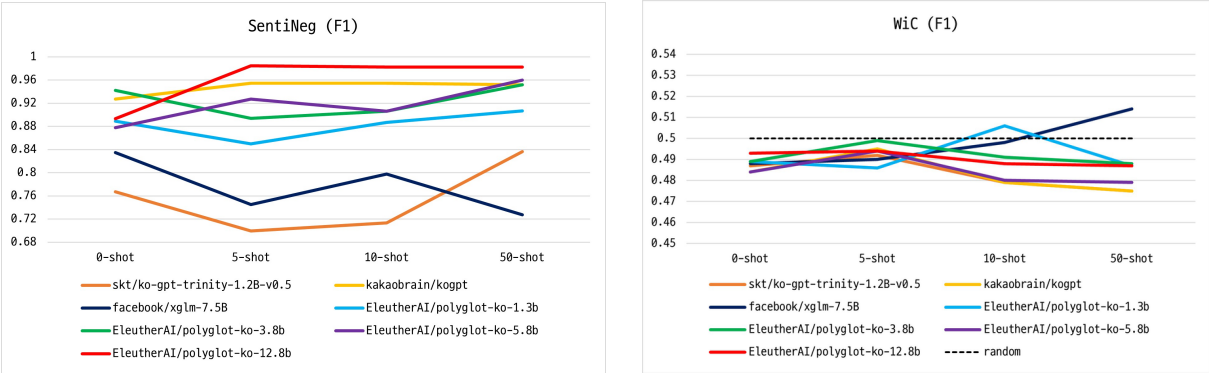

Figure 3: The performance metrics of the SentiNeg task with a modified prompt (left) and WiC task (right) using the KOBEST dataset, with the F1 scores for SentiNeg while the accuracy for WiC to show random performance.